

Alzheimer's Disease Stage Classification Using MRI Brain Images - A Comparative Analysis: Logistic Regression vs. Convolutional Neural Network

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April 2026

Abstract- This research aims to understand whether it is possible to use machine learning and deep learning models to identify the four stages of Alzheimer's disease using MRIs of brain scans. The project uses the dataset, Alzheimer-MRI, available on the platform, Hugging Face (Obscure-Entropy/Alzheimer-MRI). This dataset contains a total of 10,496 images belonging to four groups: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Demented. Comparing two approaches, we find Logistic Regression with PCA, having 78.81% accuracy rate on validation set and 78.67% accuracy rate on test set after transforming the input feature size of 16,384 into 372 principal components and a Convolutional Neural Network, having an accuracy rate of 97.17% on validation set and 96.25% on the test set. Also, the model gave a macro-F1 score of 0.96. In extensive exploratory data analysis, it was found that there exists a high-level of imbalance of 41:1 between Non-Demented images and Moderate Demented Images and that Non-Demented and Very Mild Demented Classes were almost identical. Augmentation increased the training data set from 4,096 to 8,192 images, resulting in CNN accuracy falling to 94.06%.

Keywords- Alzheimer's disease, MRI classification, Convolutional Neural Network, Logistic Regression, Principal Component Analysis, deep learning, medical image analysis, class imbalance, experimental setup, cross-validation.

GitHub:<https://github.com/Newinacode/ML-Alzheimer-disease>

I. INTRODUCTION

Alzheimer's disease (AD) is a progressive brain disease that leads to memory impairment, cognitive deterioration, and disability [2] [3]. It is essential to diagnose the disease early and accurately because early treatment can slow down its progression to a considerable extent. This research examines the potential of using machine learning and deep learning techniques for classifying four different stages of Alzheimer's disease based on MRI brain scans. This paper compares the Logistic Regression model with PCA and the Convolutional Neural Network model.

For this project, we picked the Alzheimer-MRI dataset from Hugging Face, specifically the one by Obscure-Entropy. We chose this dataset because it has a wide range of real MRI brain images. The Alzheimer-MRI dataset has 10,496 MRI images, which are divided into four groups: people who are not

demented, those who are very mildly demented, mildly demented, and moderately demented. Each image is 128x128 pixels, and since each pixel is a unique feature, that means each image has 16,384 features. This makes the dataset quite detailed and useful for our purposes. We can use these features to learn more about the differences between the images and how they relate to the different stages of dementia.

A. Exploratory Data Analysis (EDA)

A detailed EDA was carried out before we built any models. We ensured that we fully understood the data, and our knowledge of the data affected how we made all modeling decisions during our project.

1) Class Distribution Analysis:

Firstly, the number of images per class for every data split was calculated. It turns out that the problem is characterized by an imbalanced class distribution, which can be considered as one of the key findings in the EDA.

TABLE I. Class Distribution Across All Splits

Class	Train	Test	Validation	Augmented
Non Demented	2,044	634	522	2,044
Very Mild Demented	1,430	459	351	1,430
Mild Demented	579	172	145	579
Moderate Demented	43	15	6	43
Total	4,096	1,280	1,024	4,096

However, the number of images in the Non-Demented class exceeds significantly that of the Moderate Demented class since there are over 2,000 images in the training set for the first case, whereas only 43 images are available for the second one, making their ratio 41:1. Obviously, a classifier built naively will perform excellently if it assigns each data point to the Non-Demented class. As far as the Moderate Demented class goes, its validation set consists of merely 6 images.

2) Sample Visualization:

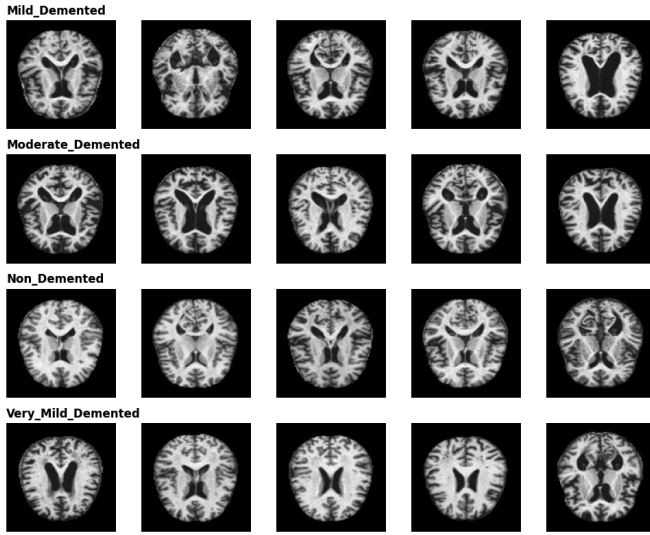


Figure 1 - 5 random sample visualization of each class

To gain more insights about our dataset, we conducted an image visualization process by displaying five random images from each category in the training dataset. As a result of the image visualization process, it was seen that the images had clear visuals and were perfectly centered. The strong finding here is that as dementia increases, the ventricle part in the middle part of the brain (black butterfly shaped object in the middle of the brain) becomes large, whereas the other parts become smaller. The most interesting aspect here is that the MRIs of Non-Demented and Very Mildly Demented people are almost alike.

3) Mean and Variance Images:

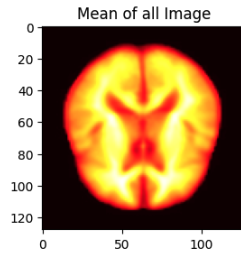


Figure 2 - Mean of training dataset

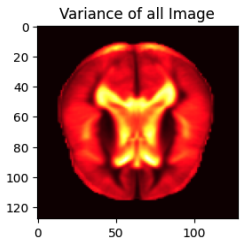


Figure 3 - Variance of training dataset

The mean image and standard deviation image were calculated separately for each class using the training data. From the

mean images, we could observe a visual difference between the Non-Demented (complete brain with less volume of ventricles) and Moderate Demented (expanded ventricles with less tissue) images. From the variance images, it was evident that the pixels around the center of the ventricle varied greatly.

II. LITERATURE REVIEW

Islam and Zhang [1] have carried out research on analyzing brain MRI images for diagnosing Alzheimer's disease using an ensemble approach with multiple deep Convolutional Neural Network models. This work is very similar to our project, as they have solved a similar four-class classification problem where MRI brain images need to be classified into Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented classes. The researchers have used the OASIS dataset to train their algorithm and have transformed three-dimensional MRI brain volumes into two-dimensional image patches with dimensions of 112×112 pixels. These patches were inputted to different CNN models with their unique Softmax layers for four-class classification [1].

There exist common features between our work and the studies conducted by Islam and Zhang. Thus, our tasks are similar, as both groups attempt to estimate the severity of Alzheimer's disease. The type of the data used by us and the authors mentioned above also appears identical: we rely on two-dimensional MR brain images. As for calculations of probabilities for each class of dementia, we and the scientists cited both employ the Softmax activation function in the final layers of our models. It should be noted that these common characteristics are very important and valuable as they reveal that different researchers may take a similar way towards addressing the same issues independently of each other [1].

There are, however, some key distinctions between their research and our research. Islam and Zhang [2] employed the OASIS dataset, while our research makes use of the Alzheimer-MRI dataset by Hugging Face, co [4] comprising 10,496 images with 128×128 pixels. More significantly, while their study only centered around deep learning with an ensemble of CNNs, our research employs a more comprehensive approach through comparing Logistic Regression with PCA and CNN approaches. No dimensionality reduction was performed in their research, but in our research, we used PCA to reduce 16,384 dimensions from the pixel-based features into 372 principal components for Logistic Regression analysis. Most importantly, there was no Exploratory Data Analysis prior to training in their study, while in our research, we conducted five distinct EDA analyses prior to any modeling, during which we identified the vital fact that there is a class imbalance ratio of 41:1 between Non-Demented and Moderate Demented images.

In conclusion, although Islam and Zhang [1] successfully showed that a CNN model could be used to classify the Alzheimer's disease stage based on MRI images. Our work is a natural extension of their research since we directly compare conventional and deep learning models using a separate public database, with a comprehensive pipeline of data analysis.

III. METHODOLOGY

In this section, we describe the process through which we developed two models that estimate the stage of Alzheimer's disease from MRI images of the brain. The two group members involved in this process were Rupesh Acharya and Prabesh Bhattarai, who developed the models based on Logistic Regression and Convolutional Neural Network respectively. The same dataset was used by the two group members for developing the two models.

A. Dataset Description

The dataset used for this project is from a particular set known as the Alzheimer-MRI dataset. The dataset was retrieved from Hugging Face. The dataset contains images of MRI brain scans amounting to a total of 10,496. The images have been divided into four distinct classes according to the severity of the disease ranging from not demented, mildly demented, moderately demented, to very demented. All the images contain 128 by 128 pixels per image hence providing 16,384 features. The images have been further divided into four subsets; namely training subset, test subset, validation subset, and augmented subset. The number of images in each subset includes 4,096 in the training subset, 1,280 in the test subset, 1,024 in the validation subset, and 4,096 in the augmented subset.

B. Logistic Regression Model

1) Preprocessing Pipeline:

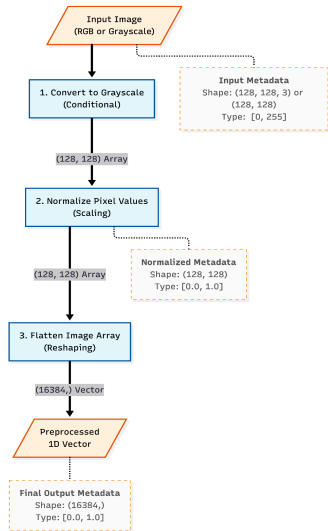


Figure 4 - Preprocessing Pipeline

These 2D matrix images were first reshaped from the dimensionality of (128,128) into a vector of 16,384 elements. All the pixel values are normalized between 0-1, by dividing each value by 255.0, since efficient gradient descent is achieved through normalization. In the augmented image dataset, images are colored and need grayscaling.

2) Dimensionality Reduction with PCA:

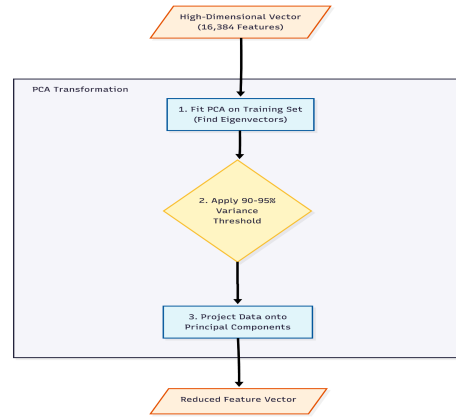


Figure 5 - PCA Transformation

The problem of overfitting was an issue because there were 16,384 features but only 4,096 training instances. The solution used for this problem was the Principal Component Analysis (PCA). PCA is a mathematical transformation method which can find the direction where there is more variation in the data and then project it on a new axis [5].

As far as the training model is concerned, we chose a variance retention threshold of 90% in which case the PCA will be required to retain the minimum possible number of principal components sufficient enough to explain at least 90% of the total variance. This resulted in 372 principal components, indicating reduction in dimensional space from 16,384 to 372. We have also carried out PCA fit with respect to our new data set, wherein our new data set consisted of both augmented and training data set values of 100% and 95% respectively, which gave us 1180 principal components. In the first method, we carried out PCA fit only with respect to our training data set and applied on validation and testing with respect to the same projection.

3) Model Configuration:

A. Logistic Regression

Now, we will take the first model in Supervised Learning, which is Logistic Regression. Logistic Regression is a classification model where you can predict the probabilities of the classes in which your inputs may fall. Despite being named "Logistic Regression," it is a classifier. Logistic regression works similarly to linear regression because in this process, you multiply each input feature with weights, add bias, and get a linear combination [6] [7]. However, in logistic regression, the output of the linear function is fed into the sigmoid/logistic function, which makes the output range from zero to one.

The sigmoid function is defined as:

$$\sigma(z) = 1 / (1 + e^{(-z)})$$

$$\text{where } z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

In this particular scenario, w refers to the weight parameters for each input parameter x , while b refers to the bias parameters. Learning here refers to the process of identifying

the optimal weight parameters via a cost function that is referred to as the cross entropy loss function. Cross entropy loss refers to the discrepancy between the actual and predicted probabilities [7] [6].

In our case, we have 4 classes i.e., Non-Demented, Very Mild Demented, Mildly Demented, and Moderately Demented. Therefore, in our case, we do not perform binary classification but rather we will be performing multiclass classification. While performing logistic regression for multiple classes, the model makes use of the softmax function rather than the sigmoid function. The softmax function takes the linear output of each class and converts it into a probability such that their sum is 1. Therefore, for each input image, our model generates a probability for each class and assigns the class with the maximum probability.

B. Model Structure

Logistic Regression - Model Flow

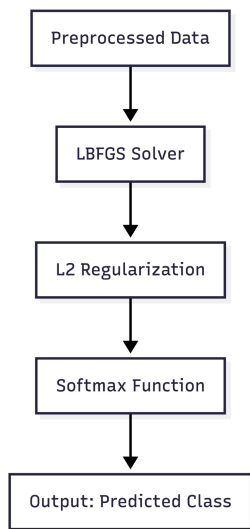


Figure 6 - LR Model Flow

Therefore, sklearn's Logistic Regression can solve multi-class problems [6]. As soon as we give our four class labels to the algorithm, it will use the multinomial method with the softmax function. The main hyperparameters that we adjust are:

solver = 'lbfgs' : This is the default solver used in sklearn. The name LBFGS means limited-memory Broyden-Fletcher-Goldfarb-Shanno. This method is a quasi-Newton technique, where an approximation of the Hessian matrix is used to determine the optimal weight. This algorithm performs better on multinomial logistic regression and relatively small data sets.

max_iter: This parameter determines the number of times that the algorithm will run in order to find the optimal weight combination. Initially, we begin by setting this parameter to a value of 1000. However, when doing experiments involving regularization, we set this to 5000.

C (Inverse Regularization Strength): The value of C is set to be 1.0 by default. The purpose of regularization is to avoid

overfitting by penalizing high values of the weight. sklearn implements regularization using the L2 method (Ridge) by default where the penalty term becomes the summation of the square of all weights. The lower the value of C, the higher is the level of regularization, while a higher value of C implies less regularization.

class_weight: The default value for this parameter is None, meaning that all classes are treated equally. But when we ran experiment two with our Model 2, we used `class_weight='balanced'`, which makes sure that the classes are weighted based on inverse frequency distribution. This means that since there are only 43 samples of class one but 2044 samples of class two, class one will be weighted more.

$\text{weight_for_class} = \text{total_samples} / (\text{number_of_classes} \times \text{samples_in_class})$

multi_class = 'auto': When auto is selected, the library is aware that there are more than two classes and thus applies the multinomial method. This indicates that it will fit one model using the softmax method that considers all classes at once.

So the overall flow for our logistic regression model is:

Input Image (128×128) → Flatten (16,384) → Normalize (0-1) → PCA (reduce features) → LogisticRegression → Predicted Class (0,1,2,3)

C. Convolutional Neural Network Model

1) Why CNN for Image Classification: The Convolutional Neural Network is a type of neural network specifically suited for processing image data. The difference between Convolutional Neural Networks and other types of models, such as Logistic Regression, consists in the way CNN handles an input. Unlike other algorithms that require flattening a two-dimensional input, the layers of CNN work with an image as it is – as a two-dimensional entity. Moreover, the filters applied by each layer scan an image looking for local patterns, such as edges, shapes and textures. Then, each subsequent layer aggregates local features into more complicated patterns, for instance, boundaries of ventricles and brain tissues patterns. Therefore, Convolutional Neural Networks would be an appropriate choice when dealing with MRI image data because of the importance of spatial coordinates [7].

2) Preprocessing: The images have been imported into NumPy arrays and standardized using the 0–255 to 0–1 scaling factor by dividing all the values by 255.0. An extra dimension has been added to each image, transforming it into an image of size (128, 128, 1), which is the desired input shape in the Keras Conv2D layers. The augmented images, which are RGB images, have been converted to grayscale.

3) Model 1 Architecture (Simple CNN — Baseline): The first model was a very basic one, consisting of two blocks of CNN, used as a base model. The Input takes the image in dimensions of (128, 128, 1). The First CNN block consists of 32 filters, 3 × 3 in size, ReLU activation, and same padding. After MaxPooling of 2 × 2 is done, the picture size changes to 64 ×

64. The second CNN block is made of 64 filters with the same attributes, after which size decreases to 32×32 . Image output is being flattened into a vector of 65,536 values. Then follows the Dense layer of 128 neurons with ReLU activation, Dropout layer with rate 0.30, and output Dense layer of four neurons with Softmax activation.

Layer (type)	Output Shape	Param #
conv2d_33 (Conv2D)	(None, 128, 128, 32)	320
max_pooling2d_33 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_34 (Conv2D)	(None, 64, 64, 64)	18,496
max_pooling2d_34 (MaxPooling2D)	(None, 32, 32, 64)	0
flatten_12 (Flatten)	(None, 65536)	0
dense_33 (Dense)	(None, 128)	8,388,736
dropout_48 (Dropout)	(None, 128)	0
dense_34 (Dense)	(None, 4)	516

Total params: 8,408,068 (32.07 MB)

Trainable params: 8,408,068 (32.07 MB)

3) *Model 2 Architecture (Improved CNN — Augmented Data)*: Model 2 represents a developed and improved variant of Model 1 that uses a bigger merged data set comprising 8,192 images. The primary improvement introduced in Model 2 as compared to Model 1 is the introduction of a third convolutional layer containing 128 filters. This leads to the gradual increase in filters in the convolutional layers by stages of 32, 64, and 128. In this way, the neural network learns to detect complex features of the brains, such as edges at level 1 (simplest), shapes at level 2, and boundaries of ventricles and tissue densities at level 3. If the use of Dropout was limited only to dense layers in Model 1, Model 2 introduces Dropout with the same rate of 0.25 for each of the convolutional layers. On the other hand, the classification layer has become even more capable owing to the increase in its size from 128 to 256 with 128 nodes and Dropout rates of 0.40 and 0.30 respectively.

Layer (type)	Output Shape	Param #
conv2d_35 (Conv2D)	(None, 128, 128, 32)	320
max_pooling2d_35 (MaxPooling2D)	(None, 64, 64, 32)	0
dropout_49 (Dropout)	(None, 64, 64, 32)	0
conv2d_36 (Conv2D)	(None, 64, 64, 64)	18,496
max_pooling2d_36 (MaxPooling2D)	(None, 32, 32, 64)	0
dropout_50 (Dropout)	(None, 32, 32, 64)	0
conv2d_37 (Conv2D)	(None, 32, 32, 128)	73,856
max_pooling2d_37 (MaxPooling2D)	(None, 16, 16, 128)	0
dropout_51 (Dropout)	(None, 16, 16, 128)	0
flatten_13 (Flatten)	(None, 32768)	0
dense_35 (Dense)	(None, 256)	8,388,864
dropout_52 (Dropout)	(None, 256)	0
dense_36 (Dense)	(None, 128)	32,896
dropout_53 (Dropout)	(None, 128)	0
dense_37 (Dense)	(None, 4)	516

Total params: 8,514,948 (32.48 MB)

Trainable params: 8,514,948 (32.48 MB)

4) Key Hyperparameters:

TABLE II. CNN Hyperparameter Comparison

Hyperparameter	Model 1	Model 2
Conv Blocks	2	3
Filter Sizes	32, 64	32, 64, 128
Kernel Size	3 x 3	3 x 3
Activation(Hidden)	ReLU	ReLU
Activation(Output)	Softmax	Softmax
Pooling	MaxPooling 2x2	MaxPooling 2x2
Dropout(Conv)	None	0.25
Dropout(Dense)	0.30	0.30, 0.40
Dense Nodes	128	256, 128
Optimizer	Adam	Adam
Learning Rate	0.001	0.001
Loss Function	Sparse Categorical Crossentropy	Sparse Categorical Crossentropy
Batch Size	32	32
Epochs	20	20

Training Samples	4,096	8,192 (train + aug)
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5) *Training Procedure*: Training for both CNNs was conducted using Adam optimization with a learning rate of 0.001 and Sparse Categorical Crossentropy as the loss function. Training was done using 20 epochs with a batch size of 32. The validation set consisted of 1,024 images to test the performance of the models and avoid overfitting. Training was done without any GPU support on local CPUs and took around 20 to 25 minutes per session. Class weighting was attempted; however, this was unstable due to the high ratio of 41:1.

IV. Experimental Setup

A. Dataset and Splits

The data set used for this experiment is the Alzheimer-MRI data set that can be found at <https://huggingface.co/datasets/Obsecure-Entropy/Alzheimer-MRI>, which consists of 10,496 MRI brain scans classified into four groups: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia. Every brain scan is of dimensions 128x128 pixels. This data set is divided into four fixed splits, as shown in below picture.

Class	Train	Test	Validation	Augmented
Non Demented	2,044	634	522	2,044
Very Mild Demented	1,430	459	351	1,430
Mild Demented	579	172	145	579
Moderate Demented	43	15	6	43
Total	4,096	1,280	1,024	4,096

B. Evaluation Metrics

Three distinct measures were utilized for assessing the effectiveness of the models. Accuracy is an indicator of how often the model has accurately predicted all the four categories of pictures. However, the macro-average F1 measure was needed due to the significant disparity between the Non-Demented and Moderate Demented categories that was 41:1. In this regard, the F1 scores of each category were calculated individually before averaging them equally. Additionally, individual measures of precision, recall, and F1 were generated for each category to identify where a particular model was performing effectively and poorly [4].

C. Implementation Details

All experiments were carried out with Python version 3.13. Logistic regression was implemented using the scikit-learn library [6] [7], whereas the CNN was built using TensorFlow with Keras. All experiments were conducted in JupyterLab

using local CPUs only. Below table summarizes the complete implementation details for both models.

TABLE III. Implementation Details

Component	Logistic Regression	CNN
Library	scikit-learn	TensorFlow / Keras
Hardware	Local CPU	Local CPU
Preprocessing	Flatten + Normalize + PCA	Normalize + Reshape
PCA Components	372 (90% variance)	Not applicable
Optimizer	lbfgs (default)	Adam (lr=0.001)
Loss Function	Not applicable	Sparse Categorical Crossentropy
Batch Size	Not applicable	32
Epochs	Not applicable	20
Max Iterations	1,000 – 5,000	Not applicable

V. RESULTS

A. Logistic Regression Results

1) *Training Performance*: Accuracy of Logistic Regression classifier on 372 features using PCA was 87.26%. But after getting a perfect F1-score in 'Moderate Demented' class (support=43) during training, it became evident that overfitting would be possible for this class because of very few data samples. The class-by-class training results are provided in below image:

Accuracy: 0.87255859375				
	precision	recall	f1-score	support
0	0.94	0.93	0.93	579
1	1.00	1.00	1.00	43
2	0.87	0.90	0.89	2044
3	0.84	0.81	0.83	1430
accuracy			0.87	4096
macro avg	0.91	0.91	0.91	4096
weighted avg	0.87	0.87	0.87	4096

2) *Validation Performance*: With respect to the 1,024 images used for testing, the accuracy reduced to 78.81%, proving that the model had been overfitted by an approximate difference of 8.45%. The class-by-class validation results are provided in below image:

TABLE IX. Logistic Regression Validation Performance

Accuracy value for Validation dataset 0.7880859375				
	precision	recall	f1-score	support
0	0.76	0.80	0.78	145
1	1.00	1.00	1.00	6
2	0.83	0.82	0.82	522
3	0.74	0.74	0.74	351
accuracy			0.79	1024
macro avg	0.83	0.84	0.83	1024
weighted avg	0.79	0.79	0.79	1024

3) *Effect of Class Weighting*: The use of `class_weight="balanced"` reduced the total validation accuracy to 76.56%, but increased the recall of mild demented patients to 0.85 from 0.80, hence demonstrating the presence of a compromise between precision and recall [4].

Accuracy value for Validation dataset 0.765625				
	precision	recall	f1-score	support
0	0.69	0.85	0.76	145
1	1.00	1.00	1.00	6
2	0.84	0.76	0.80	522
3	0.71	0.74	0.72	351
accuracy			0.77	1024
macro avg	0.81	0.84	0.82	1024
weighted avg	0.77	0.77	0.77	1024

4) *Effect of Augmented Data*: Through combination of the augmented split and the training set, 8,192 examples were achieved. PCA was performed again with 200 components. The result was a drastic reduction of accuracy to 64.45%. Optimization of hyperparameters was done for $C \in \{0.01, 0.1, 1, 10, 100\}$, and the highest value of macro F1 score obtained was 0.66 when $C = 0.1$.

Accuracy value for Validation dataset 0.64453125				
	precision	recall	f1-score	support
0	0.51	0.34	0.40	145
1	1.00	0.83	0.91	6
2	0.73	0.77	0.75	522
3	0.56	0.58	0.57	351
accuracy			0.64	1024
macro avg	0.70	0.63	0.66	1024
weighted avg	0.64	0.64	0.64	1024

5) *Test Performance*: The two Logistic Regression models which showed better results compared to other models were selected for evaluation using the test dataset which consisted of 1,280 images. The first model was selected as it achieved the maximum accuracy during the validation process, which was 78.81%. Model 3 was selected as it was trained using a different method on the augmented dataset consisting of 8,192 images.

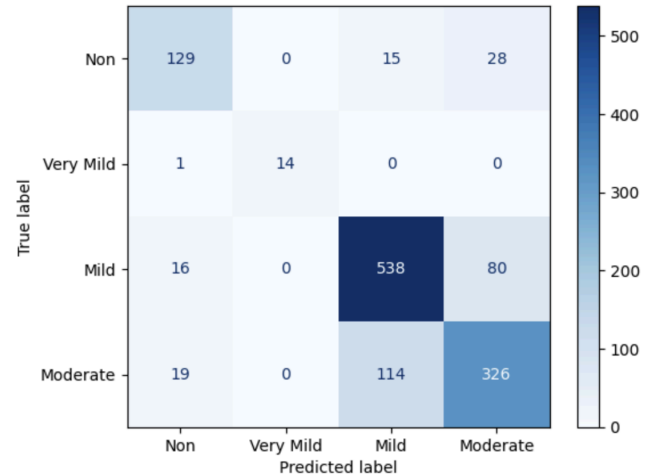
TABLE IV. Test Performance

Class	Model 1 F1	Model 3 F1	Support

Mild Demented	0.77	0.42	172
Moderate Demented	0.97	0.93	15
Non Demented	0.83	0.73	634
Very Mild Demented	0.73	0.55	459
Overall Accuracy	78.67%	63.98%	1280
Macro F1	0.82	0.66	1280

In every single case, Model 1 outperformed Model 3 by achieving 78.67% in terms of test accuracy when compared to Model 3 which recorded 63.98% test accuracy. Both Model 1 and Model 3 had Moderate Demented as their easily classified class with accuracy of 0.97 and 0.93 respectively. This happened due to the enlargement of the ventricles characteristic of that particular class. The class with the least F1 score for both models is Very Mild Demented with an accuracy of 0.73 and 0.55 respectively.

6) Confusion Matrix Analysis:

**Figure 7 - LR Model 1 Confusion Matrix on Test Data**

From the confusion matrix formed from the logistic regression model, it is evident that there are specific trends in terms of how errors have been classified in the process. For example, in case of model 1, Moderate Demented is the category that has been classified accurately because of the 14 out of 15 correct predictions, meaning a recall of 0.93. This is expected because, for images of demented brains, ventricles are significantly large, making it easier for them to be detected through a linear classifier.

The most common mistakes were found in the categorization of the Very Mild Demented category, where 326 out of 459 images were categorized accurately, resulting in a recall rate of 0.71. Most of these mistakes occurred in images categorized as Non-Demented images, thereby confirming the results of the EDA showing that both images are visually alike with nearly equal pixel intensity distribution. The other category that had

many mistakes was the Mild Demented category, with a recall rate of 0.75.

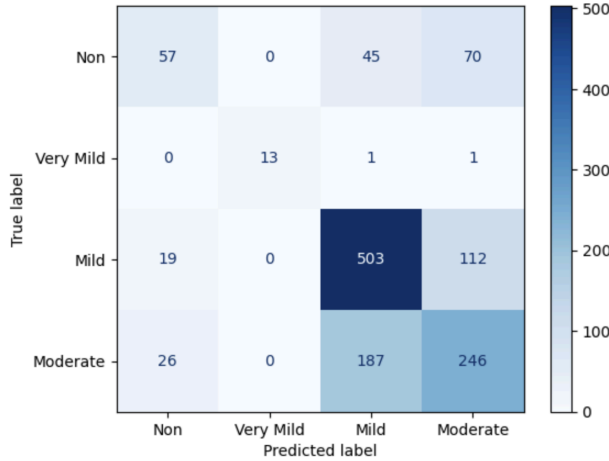


Figure 8 - LR Model 2 Confusion Matrix on Test Data

When it comes to Model 2 that was trained on augmented data, there has been a drastic fall in accuracy for all categories. There was a significant drop in recall for Mild Demented from 0.75 to 0.33, implying that only 57 of 172 samples from this category could be correctly identified. For Very Mild Demented, too, there has been a steep decline from 0.71 to 0.54. However, there is only one category of Moderate Demented, wherein only 13 out of 15 samples have been accurately predicted.

B. CNN Results

1) *Model 1 — Validation Performance:* The CNN Model 1, consisting of two blocks trained on 4,096 grayscale images, achieved 97.17% validation accuracy with a macro F1 score of 0.98. The breakdown of the class-wise validation accuracy is shown in the image below:.

TABLE X. CNN Model 1 Validation Performance

Accuracy: 0.9716796875				
	precision	recall	f1-score	support
Mild Demented	0.94	0.99	0.96	145
Moderate Demented	1.00	1.00	1.00	6
Non Demented	0.97	0.98	0.97	522
Very Mild Demented	0.99	0.96	0.97	351
accuracy			0.97	1024
macro avg	0.97	0.98	0.98	1024
weighted avg	0.97	0.97	0.97	1024

The first model achieved excellent results in terms of accuracy for all classes. The class called "Moderate Demented" showed an outstanding F1 metric, equaling 1.00, but it is necessary to be careful about this metric because the sample size of this class only consisted of six objects. All other classes demonstrated F1 scores more than 0.96 [5] [6].

2) *Model 2 — Validation Performance (Augmented Data):*

Accuracy: 0.9443359375				
	precision	recall	f1-score	support
Mild Demented	0.96	0.87	0.91	145
Moderate Demented	1.00	0.67	0.80	6
Non Demented	0.96	0.96	0.96	522
Very Mild Demented	0.91	0.95	0.93	351
accuracy			0.94	1024
macro avg	0.96	0.86	0.90	1024
weighted avg	0.95	0.94	0.94	1024

Nonetheless, despite the fact that Model 2 employed a larger number of training data points and also had a deeper neural network structure, Model 2 could not outperform Model 1 when it came to accuracy, whose level was found to be 94.43% in comparison with 97.17%. It is worth noting that the recall score for the category of Moderate Demented reduced from 1.00 to 0.67 in Model 2.

3) *Training Curves:* The training curve shows the accuracy and loss of the model after each epoch. If the two curves are moving together, it indicates that learning is happening. If they are not, it is because of overfitting.

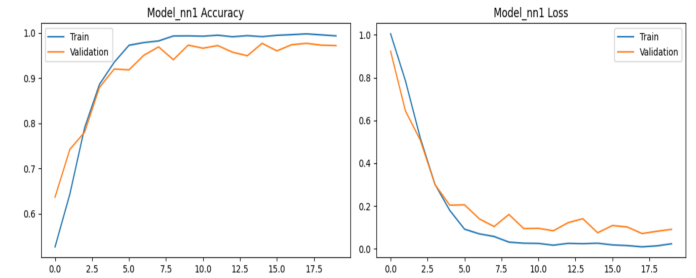


Figure 9 - Accuracy and Loss of model 1

In the case of Model 1, there is gradual improvement in both training and validation accuracy throughout the 20 epochs, whereby training accuracy stood at 99.80% and validation accuracy 97.17%. In the plot for losses, both losses were reduced up to the 10th epoch; however, after the 10th epoch, there is fluctuation in the validation loss as compared to the training loss, which continues to decrease.

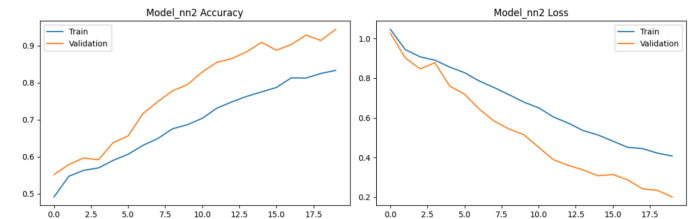


Figure 10 - Accuracy and Loss of model 2

For Model 2, the training accuracy similarly increased gradually; however, the difference between the training accuracy and the validation accuracy was greater when compared to Model 1. Even though Model 2 applied better regularization with added dropout layers following every convolutional layer, the use of more training data led to distributional changes, thus making it difficult for the model to

apply its learned parameters on the validation dataset. This is why the validation accuracy of Model 2 remained at 94.43%.

4) *Test Performance:* The two best-performing CNN models were chosen for evaluation on the test dataset, which consists of 1,280 unseen images. Model 1 was chosen since it showed the maximum validation accuracy of 97.17%. On the other hand, Model 2 was chosen in order to verify whether augmenting the dataset will help improve generalization in real-world scenarios. The test dataset was only used in this step to avoid any information leak while developing the model. Below Images presents the complete test results for both models.

Model_nn1 on test data
Accuracy: 0.9625

	precision	recall	f1-score	support
Mild Demented	0.96	0.95	0.96	172
Moderate Demented	1.00	0.93	0.97	15
Non Demented	0.95	0.98	0.97	634
Very Mild Demented	0.98	0.94	0.96	459
accuracy			0.96	1280
macro avg	0.97	0.95	0.96	1280
weighted avg	0.96	0.96	0.96	1280

Model_nn2 on test data
Accuracy: 0.940625

	precision	recall	f1-score	support
Mild Demented	0.98	0.86	0.92	172
Moderate Demented	1.00	0.80	0.89	15
Non Demented	0.96	0.96	0.96	634
Very Mild Demented	0.91	0.94	0.92	459
accuracy			0.94	1280
macro avg	0.96	0.89	0.92	1280
weighted avg	0.94	0.94	0.94	1280

The accuracy percentage of Model 1 was 96.25%, while the macro-F1 score was 0.96. This performed better than Model 2, whose accuracy percentage was 94.06%. The accuracy percentages achieved through the test data set were quite close to those achieved by the validation data set in terms of both the models, implying that neither model overfitted the validation data set. The highest variation between both models came in the case of the Moderate Demented category, wherein Model 1 achieved an F1 score of 0.97, whereas Model 2 scored 0.89.

5) Confusion Matrix Analysis:

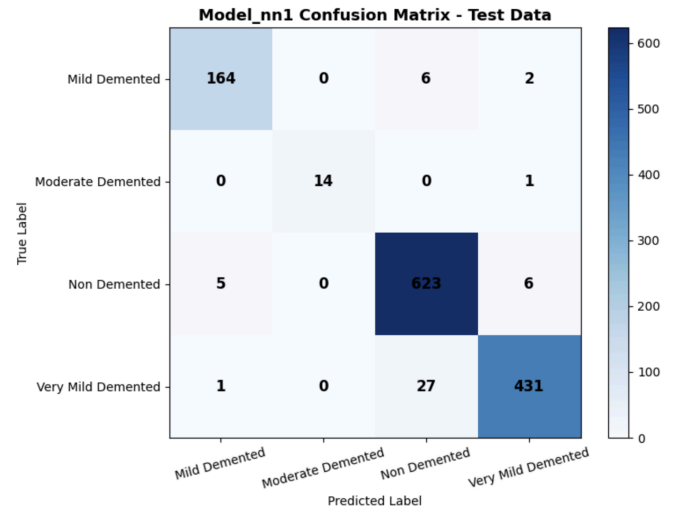


Figure 11 - Model 1 Confusion Matrix

The confusion matrix contains the exact count of correct and incorrect predictions for each individual class using the 1,280 test images. The diagonal elements correspond to correct predictions, whereas the off-diagonal elements indicate incorrect predictions.

In the case of the CNN model 1, the Mild demented category had an accuracy score of 164 out of 172 while the performance of the Moderate demented category was excellent with 14 out of 15 correct. The category of Non demented had an accuracy of 623 out of 634 while Very Mild demented had an accuracy of 431 out of 459.

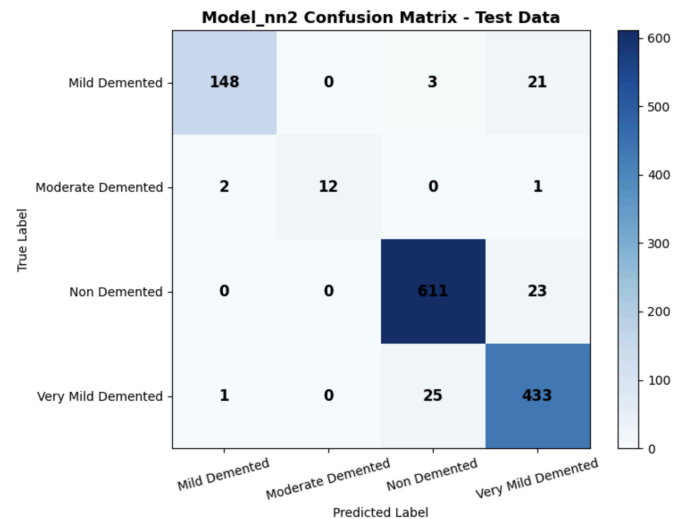


Figure 12 - Model 2 Confusion Matrix

The accuracy rate of CNN Model 3 has gone down for all classes, except Very Mild Demented. For the Mild Demented class, there are 148 correct predictions out of 172, where 21 instances were misclassified as Very Mild Demented. The accuracy rate for Moderate Demented class is 12/15. The accuracy rate of the Non Demented class is 611/634 and the accuracy rate of Very Mild Demented is 433/459.

The main observation obtained from both classifiers is that the maximum error rate occurs for the confusion of the Very Mild Demented and Non Demented groups, which agrees with the results from EDA where the two groups look alike. The Moderate Demented group is still the best classifier in both models since they have a distinctively enlarged ventricle structure.

C. Overall Comparison

TABLE V. Overall Comparison

Metric	LR Model 1	LR Model 2	CNN Model 1	CNN Model 2
Validation Accuracy	78.81%	64.45%	97.17%	94.43%
Test Accuracy	78.67%	63.98%	96.25%	94.06%
Macro F1 (Test)	0.82	0.66	0.96	0.92
Best Class	Moderate	Moderate	Moderate	Moderate
Worst Class	Very Mild	Mild	Very Mild	Very Mild
Training Samples	4,096	8,192	4,096	8,192

The most accurate results were obtained from CNN Model 1, whose accuracy rate was 96.25%, while its macro F1 score was 0.96; that is, the performance of CNN Model 1 was 17% higher compared to the most effective logistic regression model. The vast difference in the results obtained clearly indicates that CNN has performed better in MRI images classification owing to the consideration of feature locations rather than pixel locations [7] [10].

In this regard, adding the augmentation process to the training set always lowered the performance of the models rather than enhancing them. As observed, LR Model 2 performed at 78.67% when not using the augmentation technique and 63.98% when using augmentation while CNN Model 3 performed at 96.25% and 94.06% respectively for the same case.

As expected from the findings of EDA, the Moderate Demented class became the most recognizable class out of the four models, because this particular class is quite different in terms of brain size with ventricular enlargements. In contrast, the Very Mild Demented class turned out to be the least recognizable class when Logistic Regression and CNN methods were used. This is because in terms of EDA, Very Mild Demented and Non Demented classes look exactly alike in their pixel representation.

VI. ANALYSIS AND DISCUSSION

A. Why CNN Outperformed Logistic Regression

The most important result of this study is the accuracy rate obtained by CNN Model 1, which is 96.25%, whereas the accuracy rate of the best-performing Logistic Regression model is 78.67%, an increase of about 17 percentage points. The reason for the performance difference is mostly due to the fact that the Logistic Regression model uses the flattened version of the image as input, with all 16,384 features. Consequently, it destroys all of the spatial relationships between the features. Moreover, it considers only linear decision boundaries that are unable to separate the images based on complex, non-linear patterns associated with different Alzheimer's stages.

B. Effect of Augmented Data

Augmented datasets always resulted in poor model performance. For example, LR Model 2 went down from 78.67% to 63.98%, whereas CNN Model 2 was reduced from 96.25% to 94.06%. This happened because the augmentations were initially saved in RGB while training images were in grayscale. While both types of datasets underwent grayscale transformations prior to mixing, some distributional changes have occurred as a result. The latter indicates the significance of image quality and accuracy matters over the number of samples used.

C. Effect of Class Imbalance

This 41:1 ratio imbalance in the classes between Non-Demented and Moderate Demented images had an effect on both of the models. In terms of Logistic Regression, using class weights led to higher recall for the minority class but led to a decline in overall accuracy to 76.56% from 78.81%. For the CNN, the use of class weights led to a completely unstable model in which the accuracy was less than 1% and thus class weights were not used in the final models [4].

D. Most and Least Difficult Classes

Moderate Demented was found to be the best class amongst all four classes because the enlargement of ventricles and decrease in the amount of brain matter made the images highly distinctive [3]. On the other hand, Very Mild Demented was always the hardest class because it looks almost similar to Non-Demented. These results are entirely in line with the five EDA analyses done before constructing the model.

VII. TECHNICAL CHALLENGES, NEXT STEPS, AND LIMITATIONS

A. Technical Challenges

There were some problems related to technology that arose during the process of developing this project. Firstly, the problem was associated with the imbalance of the dataset that had a ratio of 41:1 between the two classes: Non-Demented and Moderate Demented classes; hence, it was unstable, considering the weight of the classes and the reduced validation accuracy of less than 1%. Also, the augmented data was saved in the RGB format, while the training data in the

grayscale format created a discrepancy in their distribution. The time needed to train the CNN on the CPU of a local system will take 20-25 minutes which is time consuming.

B. Next Steps

With one more month, here are the improvements that can be made. Firstly, we would use transfer learning by leveraging the power of pre-trained models such as ResNet50 or VGG16 since they contain features learnt from millions of images, and these features would definitely make our model perform better with a lesser amount of data [7]. Secondly, we would use k-fold cross-validation for both the models to get accurate and robust results, especially with respect to the Logistic Regression model, since it is computationally affordable. Thirdly, we would use SMOTE or Synthetic Minority Over-sampling Technique to create a few more synthetic images in the moderate demented class in order to address the issue of imbalanced classes without disturbing the model's stability [4] [8].

C. What the Model is Good For and Bad For

It has been noted that CNN has performed exceedingly well in categorizing such images having unique structures in different categories, especially in the case of the Moderately Demented class where large ventricles and small brain tissues could be observed. What comes to be difficult is that CNN does not perform well in classifying between Very Mild Demented and Non-Demented groups since both categories are similar to each other. Logistic regression, though not very accurate, is simpler in understanding and implementation than CNN.

D. Limitations

This is due to the fact that the data was collected from only one repository of Hugging Face and since there is no knowledge regarding the data collection sources, it would be unreasonable to generalize this model for other hospitals, geographical areas, or even other countries. The patient demographics, the specifications of the MRIs, and the protocols of imaging also differ greatly in other places apart from this. A machine learning algorithm trained with this particular dataset may not work well when used for scans taken in completely different environments. Moreover, since there are only 43 images available in the Training set for the Moderate Demented category, evaluating the accuracy of this model for this particular group becomes statistically unreliable. Finally, the diagnosis of Alzheimer's disease is based not only on the brain scan but also on various tests performed by clinicians.

VIII. CONCLUSION

In summary, it can be stated that the Convolutional Neural Network outperformed the Logistic Regression Model using Principal Component Analysis in distinguishing the stages of Alzheimer's disease from brain Magnetic Resonance Imaging. The CNN Model 1 achieved an accuracy score of 96.25% when trained with two blocks using 4,096 grayscale images, which is far superior to the accuracy rate of 78.67% attained by Logistic Regression Models.

extreme imbalance between classes by a factor of 41:1 and demonstrated that there is little visible difference between

classes Non-Demented and Very Mild Demented, which was the reason behind their incorrect classification in both models. An unexpected yet significant result was that the addition of extra data led to performance drop in both cases, which indicates that high quality of data is more desirable than its quantity.

In conclusion, this experiment demonstrates that CNN-based deep learning algorithms have significant promise in being used for fully automatic Alzheimer's disease staging using MRI brain imaging. Further work including the use of transfer learning, better augmentation techniques, and multi-center data testing will likely result in highly effective decision support systems for diagnosis of Alzheimer's disease [9] [6] [7].

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AUTHOR CONTRIBUTIONS

1st Author (Prabesh Bhattarai): Methodology, Experimentation, Formal Analysis, Writing Neural Network Model (CNN), Exploratory Data Analysis, Results Analysis.

2nd Author (Rupesh Acharya): Methodology, Experimentation, Formal Analysis, Writing Logistic Regression Model, Dimensionality Reduction (PCA), Results Analysis.